HERIOT-WATT UNIVERSITY

F79MB Statistical Models B **Project 1 – 2026**

Your report for this project should be submitted through Turnitin by **3:30pm (local time at your campus) on Monday 23rd February 2026**. All required data set and a link to the submission page are available through "Project 1" in Modules on the course Canvas page. Please use the submission link provided to submit your work.

This project will account for **40%** of your grade for the course.

The total length of your report should be no more than 7 pages (11-point font, A4 size), including graphs and tables but excluding any appendices. Project title and student identification information should be in page 1, not a separate title page, and count towards the 7-page limit. **You must complete the Standard Declaration of Student Authorship quiz on Canvas before you can access Project 1 questions. Please submit the report in pdf format.**

Questions for the Lecturers: If you have any questions on the assignment, please ask in the discussion board rather than e-mailing lecturers directly. That way all students will see the same responses. The discussion board will be closed at 10:00 am UK time = 1:00 pm UAE time = 5:00 pm Malaysia time on Sunday, **22nd February 2026**.

Students are allowed to discuss the methods used with other students, but your submitted project must be all your own work. **Plagiarism** is a serious academic offense and carries a range of penalties, some very serious – see Appendix A.

- Answer all tasks in this project, making sure that you add appropriate context and that your answers have a clear and logical structure and are well presented.
- You should explain carefully your work for each task so that your work demonstrates understanding of the methodology and computations that you use.
- You should include clearly labelled and correctly referenced graphs and add appropriate comments.
- You should use R to perform the required analyses and produce suitable graphs. Unless otherwise stated you do not need to explain the R commands that you use. However, you must put the R code in appropriate appendices at the end of your report and should include appropriate commenting within your R code.
- See Appendix B for the marks allocated to overall exposition/presentation.
- **Late project submissions will be penalised according to the University Policy on Submission of Coursework. That is, work submitted after the deadline but within 5 working days will be subject to a 30% deduction from the mark awarded; work submitted more than 5 working days after the deadline will be awarded a mark of zero. No individual extensions are permitted. In the case where a student submits coursework up to five working days late and has valid mitigating circumstances, the mitigating circumstances policy will apply. You can expect to receive feedback on your work by 16th March 2026.**

Tasks:

1. A large software company wants to improve its **bug management system** by better understanding the time required to resolve reported software bugs. The data set in the **bug_time.txt** file, available on Canvas under **Modules > Project 1**, consists of the **bug-fix resolution time (in minutes)** for a random sample of **200 software bugs** reported in the United States for the year **2022**.

   You can load the data into R using a command similar to

   ```
   scan("C:/R/bug_time.txt").
   ```

   Note that you will need to specify your own path to the file.

   a) Present a histogram and a boxplot of the data, as well as appropriate numerical summaries. Comment on the distribution of the data.

   [3 marks]

   b) It is suggested that the natural logarithm of the bug-fix resolution time may be modelled by a normal distribution with parameters μ and σ^2, where both parameters are estimated from the data using maximum likelihood estimation (MLE)**.**

   (i) Derive the MLEs of μ and σ^2 for the log-transformed data. Clearly show the steps involved in the derivation.

   [2 marks]

   (ii) Using the given data, compute the numerical MLEs of μ and σ^2. Compare these estimates with the sample mean and sample variance of the log-transformed data, and comment briefly on any differences observed.

   [3 marks]

   c) Use an appropriate Q–Q plot to assess whether the log-transformed data can reasonably be modelled by a normal distribution. Comment on your plot.

   [2 marks]

   d) It is suggested that the bug-fix resolution times may be modelled by a lognormal distribution. Using the estimated parameters from part 1(b)(ii), perform a χ^2 goodness-of-fit test with 10 cells, starting from zero, taking care to minimize any arbitrariness in the procedure. Clearly state the null and alternative hypotheses, and comment on your results.

   [4 marks]

2. The data set in the **app_session_duration.txt** file, available on Canvas under **Modules > Project 1**, consists of the **duration of user sessions (in minutes)** recorded for a mobile application in the United Kingdom from a **random sample of 32 sessions** during the year **2025**.

a) Present appropriate numerical summaries and graphical displays of the data, and comment on the main features of the distribution. Discuss how these features could influence your choice of statistical methods for inference.

[4 marks]

b) Use a non-parametric bootstrap methodology to obtain the empirical sampling distribution of the sample median session duration. Present this distribution graphically and comment on its shape.

Next, compute a 95% non-parametric bootstrap confidence interval for the population median session duration.

[4 marks]

c) Assume that the app session duration data can be reasonably modelled by a Lognormal distribution with parameters μ and σ^2, estimated from the data. An analyst also claims that the population median app session duration is 15 minutes or less.

Using asymptotic theory, obtain a 95% confidence interval for the population median session duration.
Discuss and comment on the validity of the confidence interval obtained here compared to the non-parametric bootstrap confidence interval obtained in 2(b).

[4 marks]

d) Next, compute a 95% non-parametric bootstrap confidence interval for the interquartile range (IQR).
Comment on the variability of the IQR relative to the median in part 2(b), and what this tells you about the session duration distribution.

[3 marks]

e) Assume that the app session duration data follows the lognormal distribution specified in part 2(c). An analyst claims that the population median app session duration is 15 minutes or less**.**

Perform a parametric bootstrap hypothesis test to assess the validity of the analyst's claim using the given dataset.

[3 marks]

[Overall exposition/presentation: 8]

[Project total: 4 0 marks]

[END OF PROJECT]

Appendix A: Plagiarism

- Coursework reports must be written in a student's own words, and any code in their coursework must be their own code. If some text or code in the coursework has been taken from other sources, these sources must be properly referenced.
- Failure to reference work that has been obtained from other sources or to copy the words and/or code of another student is plagiarism and if detected, this will be reported to the School's Discipline Committee. If a student is found guilty of plagiarism, the penalty could involve voiding the course.
- Students must never give hard or soft (electronic) copies of their coursework reports or code to another student. Students must always refuse any request from another student for a copy of their report and/or code.
- Sharing a coursework report and/or code with another student is collusion, and if detected, this will be reported to the School's Discipline Committee. If found guilty of collusion, the penalty could involve voiding the course.
- See https://www.hw.ac.uk/uk/students/studies/examinations/plagiarism.htm for more details.

Appendix B: Rubric for marks allocated to overall exposition/presentation

The 8 marks available for the exposition/overall presentation of your report will be awarded according to the scale below.

2 marks	• Some answers lack of clarity and logical structure • Some analyses may not relevant • Statements of conclusions are present but may not be very clear • Some statistical calculations and methodology set out for the reader • Some tables and figures relevant and referenced
4 marks	• Answers sometimes clearly and logically structured • Focus on key points but may include some superfluous/irrelevant analyses • Statements of conclusions generally suitable for a non-statistician • Statistical calculations and methodology are generally presented okay for the reader • Tables and figures mostly well chosen, clear, and correctly referenced • Some sources are clearly and correctly referenced, with R code included
6 marks	• Answers are mostly clear and logically structured • Clear focus on key points, avoiding superfluous/irrelevant analyses • Statements of conclusions suitable (wherever possible) for a non-statistician • Statistical calculations and methodology set out clearly for the reader • Tables and figures well chosen, clear, correctly referenced and easy to interpret • Sources used clearly and correctly referenced • R code included with comments
8 marks	• As in the previous category, answers are exceptionally clear, logically structured, and focused on key points, with no superfluous graphs, comments, or inappropriate explanations. The work demonstrates mathematical sophistication, insight, and original reasoning. R code is included where appropriate, well-commented, and concise, with no unnecessary or redundant coding.